

From Set Prediction to Evidence Selection: An Empirical Study of Surrogate Objectives in Multi-hop RAG

Abstract

Intelligent information access systems select evidence with learned scores, although users ultimately receive a reader’s answer. We examine the correspondence between an annotation-supervised objective, actual evidence selection, and downstream answers in budgeted multi-hop retrieval-augmented generation. First-, second-, and fourth-order factorization models and a DeepSets control share dense candidates, supervision, and a fixed reader. Despite high constructed-set discrimination, static fourth-order selection retains complete annotated support on 70.9% of HotpotQA development questions, below a feasible dense reference’s 82.1%; that reference is explicitly eligible for final selection. Search-aligned continuation supervises deployed scores on shared final and partial search states, with an equal-update static replay control. On the same 128-question HotpotQA/MuSiQue panel, fourth-order support completeness rises from 55 to 57 questions while answer F1 falls from 0.4181 to 0.3766. Across both examined seeds, the largest negative contribution occurs among questions retaining complete support; newly complete-support groups contribute positively. Second-order continuation produces small gains over its own static models, while maximal marginal relevance remains competitive. Exact ball indexes preserve flat-search choices but increase measured selection time at 128 candidates. These development observations motivate feasible-reference checks, paired support–answer analysis, and separate measurement of search fidelity and cost. They do not establish independent generalization or a causal effect of support completeness on answers.

Keywords: Retrieval-augmented generation , Neural set selection , Surrogate objectives , Multi-hop question answering , Search-aligned learning , Empirical evaluation

1 Introduction

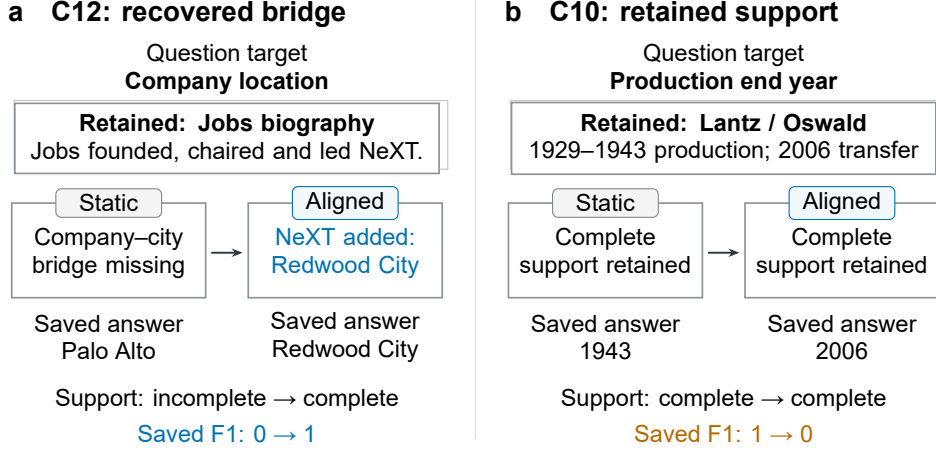
In retrieval-augmented information access, a system chooses evidence before a reader produces the user-visible answer. Multi-hop questions make this a decision over combinations of passages: individually relevant items may not supply a complete chain. A learned set scorer can recognize a supplied combination containing annotated support yet prefer another combination during search. The resulting context then informs a separate decision by the reader. This study examines these interfaces when support

annotations provide the training objective but answer quality remains the downstream endpoint.

SetR formulates context construction as set selection [6]; RECOMP learns compression and selective augmentation [16]; and evidence-chain retrieval models interactions between passages [4, 20]. These approaches move beyond independent relevance ranking. However, aggregate accuracy on supplied evidence sets does not answer whether a score preserves support when it is optimized over many combinations. Nor does an improvement in the annotation target establish an improvement in reader answers. Support annotations can omit valid alternative evidence and do not directly describe the reader’s response to a context.

We study compact neural scorers over frozen dense embeddings under a shared candidate pool, paragraph renderer, reader, and input ceiling. The models differ in interaction structure: a first-order model (H1), second- and fourth-order factorization models (H2 and H4), and a DeepSets control. All learn from complete, incomplete, replaced, and redundant support sets. A search-aligned continuation then supervises the score used for selection on a shared pool of search-generated sets, including final decisions and partial frontiers. Static replay controls the number of additional optimizer updates.

Two inspected development examples illustrate why support and answers require separate evaluation (Figure 1). In C12, adding the company-to-city bridge accompanies a change to the correct city. In C10, both the creator link and production dates remain available, yet the answer changes to a rights-transfer year already present in the source. These observations motivate paired evaluation without identifying the cause of either answer change.



Annotated-support completeness and reader utility are distinct evaluation endpoints.

Fig. 1 Two observed development examples motivating separate evaluation of evidence selection and answer quality. Short source paraphrases show H4 static and search-aligned contexts, not complete prompts or verbatim quotations. In inspected case C12 (HotpotQA, seed 2026), a recovered company-to-city bridge accompanies increased saved F1. In C10 (MuSiQue, seed 1729), complete annotated support is retained, but the answer changes from the production endpoint to a rights-transfer year present in both contexts. Arrows mark observed selection transitions. These illustrative cases neither estimate prevalence nor establish causality

Three research questions organize the comparisons. **RQ1:** Do scores that distinguish constructed sets retain annotated support when a feasible dense reference is available? **RQ2:** How does training on actual search states change support selection and answers on the same questions? **RQ3:** Can exact indexing preserve those choices at lower measured cost? The first two questions address the learned objective and its downstream consequences. The third provides a supporting implementation check, comparing vectorized Flat search with granular-ball and k-means indexes in the conditional scoring space.

The contribution is a linked empirical comparison of these decision interfaces, not a new retrieval architecture. First, an explicitly eligible dense reference separates candidate availability from score-induced selection errors. Second, same-question records distinguish changes in annotated support from changes in answers: H4’s largest negative F1 contribution occurs in the support-preserving group, whereas newly complete-support groups contribute positively. H2’s small gains over its own static models under both seeds and the competitive MMR baseline prevent a single higher-order explanation. Third, an exact-index comparison separates decision preservation from measured acceleration. The inspected source-text cases illustrate distinctions that aggregate support counts leave unresolved; they do not estimate their prevalence.

These comparisons use exposed development material: the initial static checkpoint selection included the answer panel, and source documents overlap development roles (Section 4). The findings characterize the tested support-supervised scorers and reader; they do not rank graph-based RAG systems in general.

2 Related work

2.1 Selecting interacting evidence

Dense passage retrieval provides a scalable relevance backbone [5], while multi-hop retrieval extends the decision beyond a single passage [15]. MMR balances relevance and redundancy [2]; coverage-based summarization provides another established formulation of joint selection [7]. These are important controls because a structured scorer must justify more than selecting several individually relevant passages.

SetR explicitly shifts RAG from ranking to set selection [6]. RECOMP learns to compress or omit retrieved context [16]. Beam Retrieval learns multi-hop retrieval through a beam [20]. Guo et al. build an inter-chunk interaction graph and retrieve evidence chains using semantic and structural relations [4]. Our study does not reproduce those systems or compare their published scores across incompatible settings. It examines a smaller family of annotation-supervised scorers, keeping their candidates and answer interface fixed while following the induced decisions.

2.2 Learning scores that will be optimized

Higher-order factorization machines use efficient ANOVA computations to model interactions [1]; DeepSets provides a permutation-invariant set architecture [18]. We adopt these established building blocks as controlled model families. Neither polynomial interactions nor permutation invariance alone establishes that a learned score preserves the desired ordering among searched sets.

Training on states encountered during search has a long history. Beam-search optimization incorporates beam errors in sequence learning [13], while conservative objective models address exploitation of learned objectives in offline optimization [11]. Our continuation applies direct preferences to saved final sets and partial frontiers, using known support rather than answer reward. It jointly changes the training examples, ranking loss, and checkpoint criterion. The comparison therefore evaluates that intervention as a whole, not the isolated causal effect of a particular loss term.

RAG-LER uses language-model supervision and confidence weighting to regulate a reranker [19]. Its supervisory object differs from our mapped support annotations. This distinction matters when interpreting failure: a support scorer need not represent reader utility, even if it predicts its annotation target well.

2.3 Retrieval quality, answer quality, and implementation cost

Information-system evaluation also asks whether an intermediate representation or metric serves its eventual use. Gerber et al. compare website embeddings through downstream tasks, strong baselines and processing time [3]; their cross-validated evaluation is broader than our development comparison. Zoralioğlu and Yalcin examine accuracy, calibration, fairness and popularity under an explicit recommender feedback process [21]. Their feedback experiment differs from our offline selection study, but illustrates why different endpoints should not be collapsed into one claim of system quality. Our narrower question follows one support-supervised score from supplied sets to chosen contexts and saved answers.

Tian et al. distinguish topical relevance from downstream utility using IR relevance judgments [10]. Their evaluation uses an IR collection and semantic answer measures, whereas ours uses mapped multi-hop supports and saved canonical answer EM/F1. The common issue is the transition from retrieved information to generated output; the label semantics and experimental interventions differ.

Center-radius maximum-inner-product bounds provide established tools for pruning search [9]. In this study, a granular ball indexes conditional scoring vectors. It does not represent an independently validated semantic evidence unit. We therefore evaluate preservation of Flat’s decisions separately from measured selection time.

3 Study design and models

Figure 2 separates offline fitting, online selection, and evaluation readouts. It depicts the study design rather than a new RAG architecture; supplied-set discrimination is a training diagnostic, while RQ1–RQ3 follow selected support, paired answer changes, and exact-index execution.

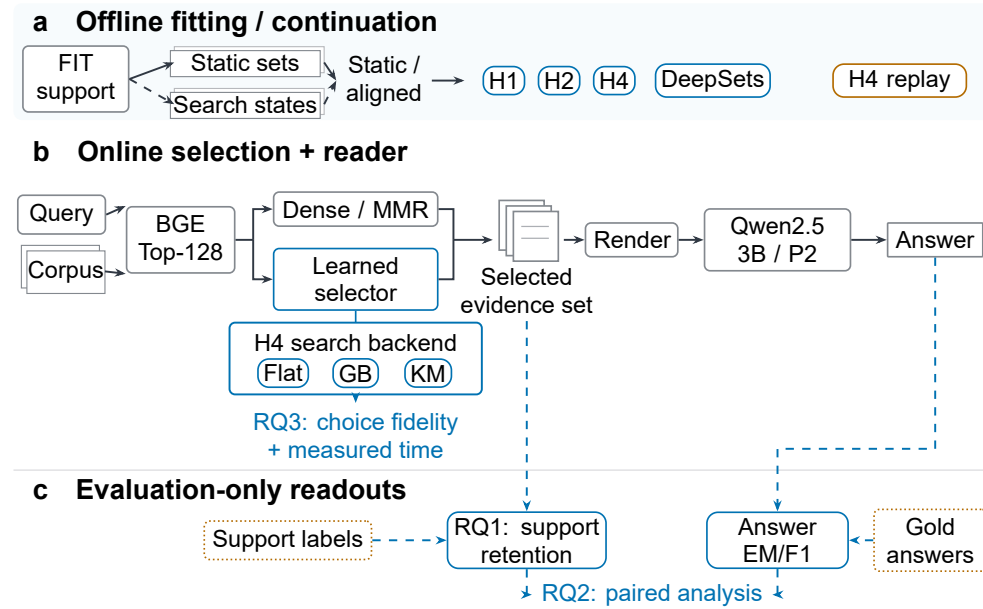


Fig. 2 Study design and evaluation interfaces. H1/H2/H4/DeepSets have separately trained parameters; static and search-generated FIT sets supply support supervision, not reader-answer reward. H4 replay matches update counts; supplied-set discrimination is a training diagnostic. With BGE frozen, all methods share the FP16 Qwen2.5-3B/P2 reader, 1,024-token serialized-input ceiling and BGE display order; only learned selectors are limited to six whole paragraphs. Feasible Dense-K6 remains eligible for their final choice. Flat, granular-ball (GB), and k-means (KM) are alternative implementations of the same H4 conditional score, not semantic-relation models or promised accelerators. RQ3 measures fidelity and time at this backend. Dashed readouts join selected support and answers in RQ2; labels never enter deployment inputs. Initial TUNE includes QA; later DEV-SELECT excludes QA. Readouts do not imply online feedback or causality

3.1 Evidence units and feasible decisions

For question q , a frozen BGE-large-en-v1.5 encoder [14] retrieves the top 128 paragraphs V_q from a per-dataset development corpus. Every selector sees this same list. Paragraphs are reconstructed from original sentence sources, with the sentence mapping retained for support evaluation. The selected whole paragraphs, including titles, are rendered in their original BGE order.

Let $T(q, S)$ denote the actual tokenizer length of the complete reader input, including question, protocol, and chat framing. Learned selection uses

$$\mathcal{F}_q = \{S \subseteq V_q : |S| \leq 6, T(q, S) \leq 1024\}. \quad (1)$$

An infeasible paragraph is skipped, not truncated to manufacture a feasible set. Tokenization of the assembled input determines feasibility; standalone paragraph lengths are features, not substitutes for this check. The six-paragraph limit and the 1,024-token ceiling are distinct constraints.

Let G_q denote known supporting annotations and $C(S)$ the annotations recovered through the source mapping. The targets are

$$\text{cov}(S) = \frac{|C(S) \cap G_q|}{|G_q|}, \quad \text{full}(S) = \mathbf{1}[G_q \subseteq C(S)]. \quad (2)$$

These labels describe annotated support. They do not mark every unannotated paragraph as irrelevant, certify factual truth, or guarantee a correct reader answer. Gold annotations are used to construct FIT supervision and evaluate saved outputs; they are not injected into evaluation candidates or supplied to the deployed scorer.

3.2 Input representation and set scores

For normalized 1,024-dimensional query and paragraph embeddings e_q, e_i , we concatenate $e_q, e_i, e_q \odot e_i, |e_q - e_i|$, cosine relevance, and log paragraph length. Scalar features use FIT normalization statistics. A linear transformation, GELU, and layer normalization produce $h_i \in \mathbb{R}^{128}$, while a query network produces h_q of the same dimension. Each model has separately trained parameters under this common input design. The features contain no query identifier, target answer, or support-membership flag.

H1, H2, and H4 have full-support and auxiliary coverage heads. For head t , the set logit is

$$g_{Ht}(S) = b_{qt}(|S|) + \frac{1}{6} \sum_{i \in S} a_{it} + \sum_{h=2}^H \frac{w_{qth}^\top E_h(S)}{32 \binom{6}{h}}, \quad (3)$$

where $H \in \{1, 2, 4\}$ is the maximum interaction order, t denotes the head, a_{it} is a unary score, b_{qt} is a query-conditioned cardinality bias, $w_{qth} \in \mathbb{R}^{32}$ contains query-conditioned order weights, and

$$E_h(S) = \sum_{i_1 < \dots < i_h, i_j \in S} v_{i_1} \odot \dots \odot v_{i_h}, \quad v_i \in \mathbb{R}^{32}. \quad (4)$$

Factors use a linear projection, layer normalization, and tanh. The denominators are fixed at the maximum cardinality six. H1 omits the interaction sum, H2 includes pair interactions, and H4 includes orders two through four. Only the full-support logit g selects sets; the coverage head is auxiliary.

The DeepSets control retains the unary and cardinality terms and adds a two-output MLP applied to $[\sum_{i \in S} h_i/6, h_q]$, with hidden width 192. It has 707,922 parameters, compared with 670,640 for H2 and 687,152 for H4. H1 registers 662,384 parameters, but 4,192 in its unused interaction projection do not affect the score. Its effective scored capacity is 658,192 parameters. The models share inputs and supervision, but this is not an exact parameter-count match.

3.3 Flat search and feasible baselines

Vectorized Flat search begins with the empty set. At each of six depths, each retained parent proposes its four highest-scoring feasible distinct extensions; the global beam keeps four unique sets. All visited sets, not only surviving parents, remain eligible for the final decision. A feasible Dense-K6 reference is also included. The final choice maximizes g , with fewer actual input tokens and then stable paragraph identities breaking ties. Negative increments do not stop expansion. Exact extension ranking does not make this finite beam globally optimal.

Dense-budget traverses the dense list and accepts each paragraph that still fits. MMR-budget uses $0.7e_q^\top e_i - 0.3 \max_{j \in S} e_i^\top e_j$, skipping infeasible additions. These two baselines may retain more than six short paragraphs. Dense-K6 and MMR-K6 add the learned selectors' cardinality limit. All methods use the same final BGE-order renderer. Consequently, comparisons with budget baselines are shared-ceiling pipeline comparisons; comparisons with K6 references more closely match selection constraints.

3.4 Exact indexes and their limited guarantee

The standard ANOVA recurrence computes the interaction terms in Eq. 3: it updates $E_h \leftarrow E_h + v_i \odot E_{h-1}$ in descending order, with $E_0 = \mathbf{1}$ [1]. For a fixed parent S , an H4 full-support extension can be expressed as

$$g(S \cup \{i\}) = o_q(S) + \theta_q(S)^\top z_i, \quad z_i = [a_i/6, v_i]. \quad (5)$$

Here $o_q(S)$ is constant across candidate additions, and $\theta_q(S)$ collects the conditional unary and interaction coefficients; the head index is omitted for readability. This converts extension ranking into maximum-inner-product search. For an index node B with center c_B and maximum Euclidean radius r_B , the standard bound is

$$\max_{i \in B} \theta^\top z_i \leq \theta^\top c_B + \|\theta\|_2 r_B. \quad (6)$$

It follows directly from Cauchy–Schwarz and $r_B = \max_i \|z_i - c_B\|_2$. An average radius would not suffice. The granular-ball index partitions by farthest pivots; the k-means index uses iterated two-center assignments. Both retain every paragraph, share the bound and token checks, and preserve ties at the pruning boundary.

The bound preserves extension ranking relative to Flat under the same score and feasibility checks. It provides no guarantee for global beam optimality or answer quality. Per-query factor construction, tree building, traversal, and token checks are included in the selection-time comparison in Section 5.3.

4 Data, supervision, and interventions

4.1 Development roles and candidate reachability

The development material contains 4,000 previously exposed HotpotQA and MuSiQue questions [12, 17]. The group split yields FIT 3,166, TUNE 738, and historical development 96 (Table 1); FIT contains 3,165 groups. The reconstructed corpora contain 9,791 HotpotQA and 31,130 MuSiQue paragraphs. This is corpus retrieval over the available development material, not full-Wikipedia retrieval. BGE retrieval is exact, without ANN approximation. Its 512-token encoder limit truncates 16 and eight paragraphs, respectively; reader feasibility is checked separately.

Table 1 Development roles and their nesting. MINE is inside FIT; DEV-SELECT and QA are disjoint subsets of TUNE. Initial static checkpoint selection used all TUNE, including QA

Role	HotpotQA	MuSiQue	Use
FIT	748	2,418	Supervised training
TUNE	196	542	Initial checkpoint loss; selection analysis
Historical DEV	56	40	Earlier exposed development
MINE (FIT)	256	256	Shared search-state pool
DEV-SELECT (TUNE)	64	64	Continuation checkpoint selection
QA (TUNE)	64	64	Saved answer comparison

The answer panel is drawn from TUNE, which was used to select the initial static checkpoints. Only checkpoint selection for the continuation excludes these questions through a separate DEV-SELECT subset. The answer comparisons therefore remain exploratory development results. The top-level FIT, TUNE, and historical-development question groups are disjoint, but FIT and TUNE share 57 HotpotQA and 4,753 MuSiQue source documents. QA contains 48 HotpotQA bridge and 16 comparison questions, and 32 two-hop and 32 longer-hop MuSiQue questions. Every model and seed reuses these 128 questions.

On TUNE, Top128 contains complete known support for 190/196 HotpotQA and 367/542 MuSiQue questions. Considering the original support paragraphs separately, known support fits the input and cardinality limits for 196/196 and 533/542. Reachability and feasibility are different bottlenecks: a support set may fit but not be retrieved. All questions, including nine budget-infeasible MuSiQue cases, remain in the unconditional results.

4.2 Static supervised sets

Static FIT supervision includes complete known-support sets, missing-support sets, same-size replacements, redundant additions, Dense-K6/MMR-K6, and limited random examples. FIT reference construction can append missing support paragraphs and matched distractors to create training sets. This training operation is not used by evaluation or search-state mining, which retain the original Top128.

The base loss combines full-support binary cross-entropy, 0.5 times sigmoid coverage MSE, 0.25 times pointwise annotation classification, and 0.25 times a softplus preference for constructed complete/incomplete pairs of equal cardinality. AdamW uses learning rate 10^{-3} and weight decay 10^{-4} . The initial study retained seeds 1729, 2026, and 426; the continuation comparison uses 1729 and 2026. The third static seed is retained in the numerical supplement but is not relabeled as a continuation run.

Constructed-pair accuracy tests a supplied pair distribution. Actual selection tests an optimized distribution of sets, including combinations never represented by those particular pairs. We therefore report pair accuracy, selected support, reader answers, and cost as distinct endpoints. The same-size training control limits an obvious cardinality shortcut but does not eliminate all length or distribution effects.

4.3 Shared search-aligned continuation

MINE contains 512 FIT questions selected by deterministic group hash, balanced across datasets with existing question-type strata retained. Each model searches with its current score. A shared pool merges final sets, high-scoring partial frontiers, Dense-K6/MMR-K6, and at most eight deterministic feasible replacement attempts per trajectory. The deduplicated pool is capped at 64 sets per question, prioritizing model finals and high-scoring frontiers. All four models train on the same pool; no model receives exclusive candidates, stronger supervision, or a different reader.

For the annotation utility $u(S) = [\text{full}(S) + \text{cov}(S)]/2$, a preferred pair has $u(S^+) > u(S^-)$. The added loss directly supervises the deployed logit:

$$\ell(S^+, S^-) = \log \left(1 + \exp \left\{ 0.2[u(S^+) - u(S^-)] - [g(S^+) - g(S^-)] \right\} \right). \quad (7)$$

Final-set pairs prefer equal cardinality and token ratios within $[0.8, 1.25]$, falling back to equal cardinality. Frontier pairs prefer a shared parent and depth, with same-depth fallback. Missing pair categories are masked. Partial sets with zero full labels can still receive different preferences through coverage, and those preferences affect the deployed score g , not merely the auxiliary head.

The continuation loss is $L_{\text{base}} + 0.5L_{\text{final}} + 0.5L_{\text{frontier}}$. Each query supplies four static and four mined base observations and up to one pair per added category. Losses are normalized within and across queries. Each of at most two rounds permits eight epochs, using AdamW at 3×10^{-4} , weight decay 10^{-4} , query batch eight, and gradient clipping at one. A second round re-mines the selected current models.

The pool does not yield a usable preference for every question. In primary-seed round one, final pairs are unavailable for 171/256 HotpotQA and 115/256 MuSiQue questions; frontier pairs are unavailable for 17 and 42. Every question still supplies base supervision. The supplementary method notes retain the corresponding round-two

counts. Thus pool size alone does not measure how many questions receive pairwise supervision.

4.4 Checkpoint and replay controls

At epochs two, four, and eight, continuation checkpoints are compared by dataset-equal complete support on DEV-SELECT, then coverage, lower original loss, and earlier checkpoint. Every selected model comes from the first round. QA-answer scores do not enter this decision. H4 static replay uses eight static sets per question and matches the selected aligned H4 checkpoint’s optimizer-update count. It does not match total training computation, because the pair terms require additional evaluations. Continuation changes the sampled sets, preference losses, and checkpoint criterion together, so the comparison evaluates the combined intervention rather than isolating a loss term.

4.5 Reader and paired outcome analysis

The reader is the original FP16 Qwen2.5-3B-Instruct [8], with greedy decoding, at most 128 output tokens, and the shared P2 single-answer-field protocol. It uses no relation-front-end adapter. Identical model, input tokens, precision, decoding, and protocol identities permit cached output reuse; cache reuse is not independent generation. Existing canonical EM and token F1, including dataset-specific answer handling, are reused without rescoring or answer cleanup. Format failures remain in the denominator.

We paired completed outputs by question, dataset, model, seed, budget, and protocol. Full-support transitions are 00, 01, 10, 11, where the first and second digits indicate completeness before and after continuation. F1 changes above 10^{-12} or below -10^{-12} define increase or decrease; the tolerance handles floating-point differences. For transition group A , its contribution to the observed panel mean is

$$d_A = \frac{1}{N} \sum_{i \in A} (F1_{i,\text{aligned}} - F1_{i,\text{static}}), \quad N = 128. \quad (8)$$

The four contributions sum to the panel difference. Unlike the within-group mean, d_A uses all 128 questions as its denominator. This is a post hoc descriptive decomposition, not a causal attribution or an inferential test. All four models and both seeds are retained, with per-dataset tables in the supplement.

We grouped saved selection logits into fixed sigmoid bins $[0, .2, \dots, 1]$ and reported counts and annotated completeness, including empty bins. These are search-selected TUNE outputs, not an independent calibration sample. For source-text analysis, we selected 16 distinct H4 questions by a fixed hash within dataset, support-transition, and F1-direction cells: primary-seed gain/harm cells first, then new questions from the second seed, with unchanged cells only if needed. Question and source text were read before saved answers. This outcome-stratified case selection does not estimate population error rates.

4.6 Reproducibility and research assistance

Online Resource 1 describes the preparation, model identities, decoding settings, data-role overlaps and cost-accounting scopes. Online Resource 2 links saved outcomes to paired tables and score bins through a standalone numerical reconstruction script and retains the existing reference and cost records. Reconstruction neither reruns a model nor rescores an answer or performs a statistical test. It must therefore be distinguished from independently regenerating the predictions. Raw source text, identifiable cohort manifests, source mappings and trained checkpoints are not supplied in this numerical package.

OpenAI’s ChatGPT and Codex assisted with research planning, implementation and debugging, analysis scripts, and source-text case interpretation. Exact historical backend versions were not retained. Numerical checks compared paired totals and endpoint means with saved records; source-text observations were recorded before consulting the corresponding predictions. This assistant-assisted case review is not independent expert annotation. The data figures are reproducibly rendered from the numerical tables with Matplotlib, using plotting code prepared with Codex assistance; no image-generation model produced experimental data or figure content.

5 Results

5.1 RQ1: Does supplied-set prediction transfer to selection?

Constructed complete/incomplete pair accuracy is high under the primary seed. H4 achieves 0.9645 on HotpotQA and 0.9464 on MuSiQue; H2 achieves 0.9638/0.9422 and DeepSets 0.9620/0.9482. H1 is lower at 0.9475/0.8978. These values show that the supplied training-style distinction is learnable, but do not establish that H4 ranks the alternatives encountered in search better than its controls.

Table 2 Complete annotated support on all TUNE questions, static H4 versus its feasible Dense-K6 reference, seed 1729. Rates are proportions; lost/gained are question counts relative to Dense-K6. Per-question reference logits were not retained

Dataset	N	Dense-K6 complete rate	H4 complete rate	Lost	Gained
HotpotQA	196	0.8214	0.7092	27	5
MuSiQue	542	0.2565	0.2232	42	24

The feasible-reference comparison makes that distinction concrete (Table 2). Dense-K6 recovers complete annotated support for 82.1% of HotpotQA TUNE questions, compared with 70.9% for static H4. Relative to the reference, H4 loses 27 complete cases and gains five. On MuSiQue it loses 42 and gains 24, with completeness 22.3% versus 25.6%. Since Dense-K6 is explicitly eligible for final selection, these errors cannot all be attributed to the beam failing to visit that feasible reference.

The reference diagnostic reports no selected H4 score below Dense-K6 (tolerance 10^{-8}). This local comparison points to the score’s ordering of feasible alternatives rather than failure to visit that reference. Only aggregate score differences and support-transition counts are available: the diagnostic did not retain per-question Dense-K6 scores or sets, and paired MMR-K6 scores are also missing. The records therefore support the aggregate comparison, but not a per-question score-gap distribution or a measure of global optimization regret.

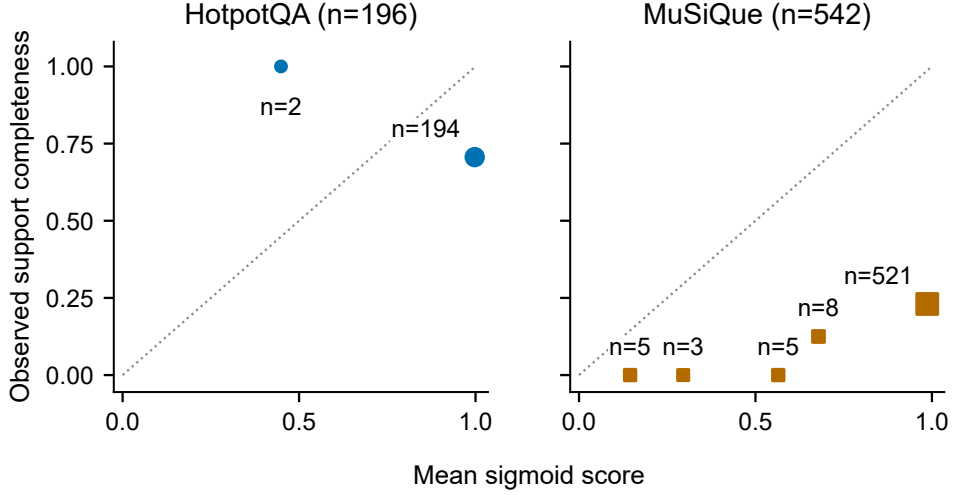


Fig. 3 Sigmoid-transformed scores and observed annotated-support completeness for selected H4 sets, seed 1729. Each point summarizes an occupied fixed bin; labels give its count, and marker area increases with the square root of that count. HotpotQA bins $[0, .2)$, $[.2, .4)$, and $[.6, .8)$ are empty; MuSiQue has no empty bins. Empty bins are not interpolated. The diagonal marks equality, not a fitted calibration curve. These selected TUNE outputs provide a descriptive comparison, not independent calibration

Figure 3 shows the discrepancy among saved selected sets directly. In the highest bin, 194 HotpotQA sets have mean sigmoid output 0.9976 but annotated completeness 0.7062. The corresponding 521 MuSiQue sets have mean 0.9867 and completeness 0.2303. The supplement gives the same bins for all four models. Concentration near one is not unique evidence of a high-order failure: the highest DeepSets bins contain 194 and 539 sets, with completeness 0.6959 and 0.2319. Search-selected outputs should therefore not be interpreted as calibrated probabilities from their numeric range alone.

A separate existing matched-length pair diagnostic also limits the higher-order interpretation. H4 pair accuracy is 0.8333/0.7878, compared with 0.9524/0.8525 for DeepSets, on 84/278 pairs from 65/185 queries. This smaller diagnostic changes the supplied-pair distribution and is not a new QA evaluation. Together, the results support a decision-distribution mismatch in the tested scorers; they do not show that adding interactions universally improves evidence choice.

5.2 RQ2: What changes after search-aligned training?

Table 3 Answers and contexts on 128 development questions (64 per dataset), seed 1729. Complete support is a question count; F1 and EM are proportions; paragraph and input-token counts are means. Dense/MMR share the 1,024-token ceiling but may exceed six paragraphs. Replay matches optimizer updates

Selector	Complete support	F1	EM	Mean paragraphs	Mean input tokens
Dense	64/128	0.4055	0.3359	6.95	1009.9
MMR	61/128	0.4303	0.3672	7.25	1007.3
H4 static	55/128	0.4181	0.3594	5.91	916.9
H4-replay	55/128	0.4046	0.3359	5.91	900.9
H1 aligned	61/128	0.4065	0.3359	4.45	696.2
H2 aligned	56/128	0.4057	0.3516	5.53	789.7
DeepSets aligned	60/128	0.4090	0.3516	4.43	693.4
H4 aligned	57/128	0.3766	0.2969	5.70	842.2

Table 3 retains simple selectors, the static H4 checkpoint, replay, and all aligned models on the same answer panel. MMR has answer F1 0.4303, compared with 0.4181 for static H4 and 0.3766 for aligned H4. H4’s complete-support count rises from 55 to 57, but EM falls from 0.3594 to 0.2969. Replay gives F1 0.4046 with 55 complete-support questions. Additional static updates therefore do not reproduce the aligned endpoint, although equal updates are not equal total compute.

The contexts differ in realized size. Dense and MMR average 6.95 and 7.25 paragraphs and about 1,010 and 1,007 input tokens. Static and aligned H4 average 5.91 and 5.70 paragraphs and 917 and 842 tokens. This is a meaningful shared-ceiling comparison but not a context-length-matched answer experiment. Shorter aligned input may reduce reader work, yet the intervention did not isolate length from content. Its F1 change cannot be attributed to shortening alone.

The response to continuation differs across models (Table 4). Relative to its own static checkpoints, H2 gains 0.0060 and 0.0051 F1 under seeds 1729 and 2026. H1 gains 0.0326 under the first seed but loses 0.0151 under the second. DeepSets and H4 lose F1 under both despite increased complete-support counts. For second-seed H4, full support changes from 53 to 57 and F1 from 0.4304 to 0.3831. The positive H2 differences and negative H4 differences are point estimates on the same development questions, without a significance or equivalence interpretation.

5.2.1 Joint support–answer transitions

Under the primary seed, full support improves for four questions and worsens for two; 122 retain their status. F1 improves for ten, worsens for 14, and remains unchanged for 104. Those margins alone cannot identify which support changes accompany answer changes. Figure 4 and Table 5 use their actual intersection.

Table 4 Continuation relative to each model’s own static checkpoint. Complete-support counts are out of 128; F1 differences use the same full panel. Both seeds reuse the same questions

Seed	Model	Complete support static $\rightarrow$ aligned	Static F1	Aligned F1	Δ F1
1729	H1	50 $\rightarrow$ 61	0.3738	0.4065	+0.0326
1729	H2	47 $\rightarrow$ 56	0.3998	0.4057	+0.0060
1729	H4	55 $\rightarrow$ 57	0.4181	0.3766	-0.0415
1729	DeepSets	54 $\rightarrow$ 60	0.4123	0.4090	-0.0033
2026	H1	49 $\rightarrow$ 56	0.4277	0.4126	-0.0151
2026	H2	53 $\rightarrow$ 54	0.4123	0.4173	+0.0051
2026	H4	53 $\rightarrow$ 57	0.4304	0.3831	-0.0473
2026	DeepSets	45 $\rightarrow$ 56	0.4027	0.3521	-0.0505

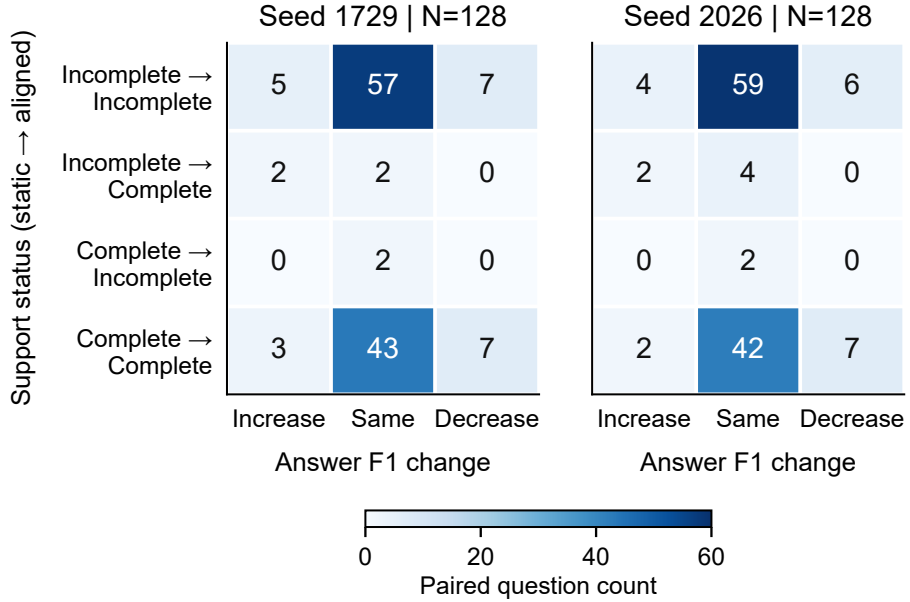


Fig. 4 Joint changes in annotated-support completeness and answer F1 for H4. Each seed uses the same 128-question panel. Cells show paired counts, including every unchanged case, under a common color scale. Support transitions and F1 directions describe associations between recorded endpoints, not causal effects

For seed 1729, questions retaining complete annotated support contribute -0.0320 to the overall F1 change of -0.0415 , while the newly complete-support group contributes $+0.0130$. The corresponding contributions for seed 2026 are -0.0477 and $+0.0091$, with a total change of -0.0473 . Every contribution uses all 128 questions as its denominator.

Table 5 H4 answer-change decomposition by annotated-support transition. Group mean divides the signed sum by group size n ; panel contribution divides it by all 128 questions. The four contributions sum to the panel difference. Here 0 and 1 mean incomplete and complete annotated support

Seed	Support transition	n	$\sum \Delta F1$	Group mean $\Delta F1$	Panel contribution
1729	0 $\rightarrow$ 0	69	-2.8810	-0.0418	-0.0225
1729	0 $\rightarrow$ 1	4	+1.6667	+0.4167	+0.0130
1729	1 $\rightarrow$ 0	2	+0.0000	+0.0000	+0.0000
1729	1 $\rightarrow$ 1	53	-4.1000	-0.0774	-0.0320
2026	0 $\rightarrow$ 0	69	-1.1190	-0.0162	-0.0087
2026	0 $\rightarrow$ 1	6	+1.1667	+0.1944	+0.0091
2026	1 $\rightarrow$ 0	2	+0.0000	+0.0000	+0.0000
2026	1 $\rightarrow$ 1	51	-6.1000	-0.1196	-0.0477

The largest negative component therefore lies in the support-preserving group in both runs. The zero contribution of the complete-to-incomplete group and the remaining negative contribution of the incomplete-to-incomplete group are shown in Table 5. These signed contributions locate the changes; they neither identify a cause nor imply that every question gaining support improved.

Full-support status is binary and intentionally coarse. Within 11, the selected surrounding paragraphs, wording, ordering under the common renderer, and actual token lengths can still change. Within 00, partial coverage can change even though completeness remains zero. The decomposition locates changes; it does not identify an internal attention mechanism or establish which context modification caused an answer to change. The numerical supplement retains coverage, EM, paragraph-count, token-count, and context-identity differences for every pair.

5.2.2 What the source-text cases clarify

The selected 16 cases were read in full using retained, removed, and added paragraphs before consulting saved predictions. Four illustrate different limits of the aggregate metrics. An added team paragraph supplies the missing conference bridge in C03; support changes 0 $\rightarrow$ 1 and F1 rises from 0 to 0.6667, while EM remains zero. C12 adds a company-location paragraph tied to the precise role in the question; support changes 0 $\rightarrow$ 1 and F1 rises from 0 to 1. These are concrete instances in which a recovered bridge accompanies an improved answer.

C10 retains the creator identity and production-date range in both contexts, but the answer changes from the requested production endpoint to a later rights-transfer year in the same source. Full support remains one while F1 drops from one to zero. The text supports a role/date-selection error description, not a claim about why the reader attended to one date. C11 retains a sponsor’s full name and acronym; the output switches from the full name to the acronym, with saved canonical F1 changing

from one to zero. That lexical change is not equivalent to losing the named entity from the evidence. We retain the original score and do not introduce a favorable alias rule.

All 32 saved answers in the 16 inspected pairs reached the end-of-sequence token before the output limit. Truncation therefore does not explain the changes in these cases. Other cases include unresolved entity links and geographic ambiguity, so a higher answer score alone does not certify a sufficient evidence chain. The supplement reports all 16 observations and their uncertainty. They illustrate plausible reading and lexical-matching errors without estimating their frequency across the full panel.

5.3 RQ3: Does exact pruning improve actual cost?

The original index check compared both indexes with Flat on 738 TUNE questions: 1,476 final comparisons and 30,956 extension states showed no mismatch. A fixed refined-leaf diagnostic used 32 questions per dataset and likewise found no mismatch across 128 final comparisons and 2,688 states. These checks assess preservation under the same score and beam, rather than global optimality.

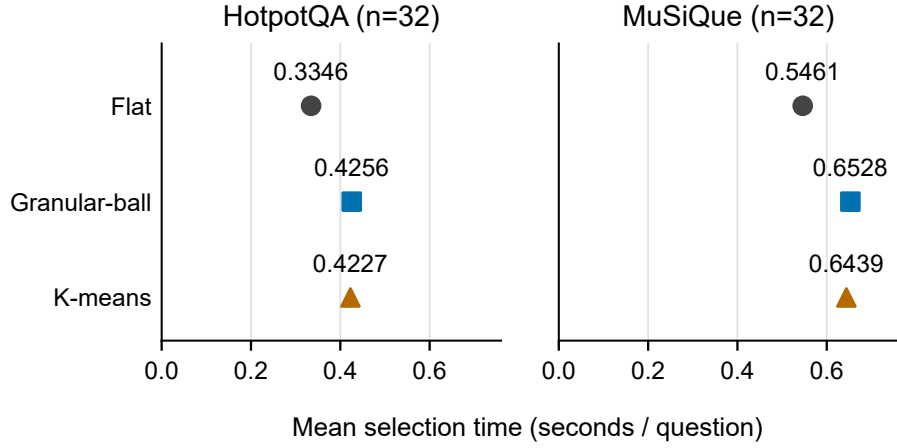


Fig. 5 Measured selection time at 128 candidates on a fixed 32-question subset per dataset. Points denote means for Flat and the two exact indexes. Timings include factor construction, per-query indexing, and token-feasibility checks. Some CPU measurements overlapped reader execution, so this is not an isolated hardware benchmark

At 128 candidates, fewer inner-product evaluations did not produce a faster selector (Figure 5). The granular-ball index increased mean selection time from 0.3346 to 0.4256 seconds per HotpotQA question and from 0.5461 to 0.6528 seconds per MuSiQue question. K-means took 0.4227 and 0.6439 seconds, respectively. The granular-ball index reduced mean true extension scores from 2,688 to about 906 and 1,505 but increased tokenizations from 215 to 281 and from 310 to 383. Pruning savings thus coexisted with more feasibility work; the measurements do not isolate either component’s causal contribution to total time.

Table 6 retains means and medians for the refined 64-question subset. These timings include factor computation, per-query index construction, and tokenizer feasibility checks. Tokenizer time is nested within total selection time and must not be added again. Full process setup and corpus encoding have different accounting scopes. Per-question timing distributions for this subset were not retained; the original 738-question timings are a separate panel in the supplement. Neither panel establishes scaling to a larger corpus or candidate count.

Table 6 Measured selection time in seconds per question, 32 questions per dataset. Means and medians are retained aggregates of the refined-leaf diagnostic; per-question timing distributions are unavailable

Selector	HotpotQA		MuSiQue	
	Mean	Median	Mean	Median
Flat	0.3346	0.1694	0.5461	0.1868
Granular-ball	0.4256	0.2781	0.6528	0.3005
K-means	0.4227	0.2632	0.6439	0.2994

The supplementary cost ledger reports model-residency GPU wall time, measured CPU time, reader calls, cache reuse, and memory for the broader completed studies. Those totals include development, sensitivity, and repeat work; dividing them by QA’s 128 questions would not estimate serving cost. Unrecorded earlier CPU remains unknown.

6 Discussion and limitations

6.1 What the induced decisions reveal

A feasible dense reference exposes an error that candidate reachability alone cannot explain: the scorer can prefer an alternative with less annotated support even when that reference reaches the final decision. Supplied-pair accuracy therefore leaves an important part of the selection problem unmeasured. The matched-length diagnostic and selected-score bins further indicate that performance depends on the distribution of sets presented to the model. An evaluation should retain an explicit feasible reference alongside constructed examples and searched outputs.

Training on search-generated sets changes the induced choices, but the model-specific responses resist a single higher-order explanation. H2 gains slightly under both seeds, H1 changes direction, and H4 and DeepSets lose answer F1 despite improved completeness counts. The common pool and replay control narrow some alternatives, yet samples, preference losses, and checkpoint criteria change together. These results motivate evaluating the combined training intervention at each decision stage rather than attributing its outcome to one loss or interaction order.

The joint analysis clarifies why aggregate support gains cannot predict the answer change. H4’s largest negative contribution comes from questions preserving complete support, while newly complete-support groups contribute positively. A binary annotation target can remain fixed despite changes in paragraph membership, length, and relative position. The cases illustrate an additional distinction between selecting the wrong date role and producing an acronym that loses canonical lexical overlap. Neither requires changing the scoring rule after seeing predictions. Reporting paired support and answer outcomes makes these differences visible.

For implementation choice, the exact indexes preserve decisions but add measured time at this scale. Correctness and fewer inner products are insufficient evidence of practical acceleration. Total measured selector time, with its accounting scope explicit, is the relevant comparison here.

6.2 Alternative explanations and study boundaries

The experiments use one frozen encoder, one reader, and one primary 1,024-token interface. Model family, realized context length, and intervention details interact. H4’s shorter aligned inputs and different paragraph membership are not isolated factors. MMR and Dense have the same token ceiling but not the same maximum paragraph count; this limits claims of a pure learned-versus-unlearned architecture comparison. The K6 diagnostic establishes a narrower feasible-reference gap in support, without supplying a new K6 answer experiment.

The data are exposed development material. Initial model selection used a TUNE pool containing QA, and source documents overlap FIT and TUNE. Separating later DEV-SELECT does not erase those dependencies. The two seeds reveal sensitivity on the same questions, not two independent confirmation samples. We retain unconditional denominators and do not use a successful-parse, reachable-only, or support-gaining subset as the main answer result. The descriptive analyses add no inferential status to the earlier records.

Support mappings may omit valid alternative evidence. An incomplete annotation set can accompany a correct answer, and complete annotated support can accompany an incorrect or differently phrased answer. Source-text review helps interpret particular cases but is assistant-assisted, outcome-stratified, and limited to 16 questions. It is neither blinded multi-annotator validation nor proof of factual correctness. Per-query Dense-K6 logits and refined-subset timing distributions are missing; the article explicitly distinguishes retained aggregates from newly reconstructible tables.

External graph and hypergraph systems were not run under the same conditions. Their published results cannot establish either an advantage or a failure of the compact scorers evaluated here.

6.3 Practical implications

For an information-access system using support-supervised selection, this study motivates three concrete checks. A *feasible-reference check* asks whether a simple reference is reachable under the same constraints and remains eligible for the scored decision; it

avoids confusing a score-ordering error with failure to retrieve the reference. A *paired-outcome check* follows the selected support and the reader answer on every question, retaining unchanged and unsuccessful cases rather than reporting only recovered-support subsets. A *measured-cost check* distinguishes exact choice preservation from the time spent constructing factors, indexing, checking tokens and selecting contexts. MMR and low-order controls are important in all three comparisons. These are lessons from the tested system, not a universally validated evaluation framework or evidence that one selector family is preferable across deployments.

7 Conclusion

This study separates recognizing annotated evidence sets from selecting contexts and using them to answer questions. A feasible dense reference exposes scoring errors that cannot be explained solely by search failing to visit that reference. Training on shared search states changes the induced decisions, but answer effects differ across model families and seeds. H2 shows small gains over its static models, while H4 and DeepSets lose answer F1 despite higher completeness counts. The joint analysis locates H4’s largest negative contribution among questions retaining complete annotated support; recovered support contributes positively. This is a description of paired outcomes, not a causal effect of completeness. Exact ball indexes preserve the choices without reducing measured time at 128 candidates. The practical implication is to evaluate learned selectors on the sets they choose, alongside downstream answers, simple controls, and measured costs. These conclusions remain limited to the exposed development material, compact scorers, and shared reader studied here. They do not establish an inherent benefit or failure of higher-order representations.

Statements and Declarations

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During manuscript preparation, OpenAI’s ChatGPT and Codex were used for literature search and organization, drafting, English revision, and cross-reference checks. The human authors retain responsibility for the article’s scientific claims and content. Research uses of these tools are described in the Methods section.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The study uses existing HotpotQA and MuSiQue materials. Online Resource 1 describes the original data releases and reconstruction prerequisites. Online Resource 2 contains de-identified saved scores, paired descriptive tables, source-role information, and a standalone reconstruction script. It reproduces numerical summaries without model inference or access to raw benchmark answers. Source texts, source identities, historical cohort manifests, and trained weights are not bundled. Release of broader derived materials remains subject to source licenses and author approval; no public weight release or complete prediction-regeneration package is claimed.

Ethics and consent

This computational study uses existing benchmark materials and involves no newly recruited participants or new collection of personal data. Ethics approval and participant consent are not applicable to this study design. The numerical resources omit raw source text and identifiable cohort records; access to the original benchmarks remains subject to their source terms.

Supplementary information

Online Resource 1 is a methods and reproduction guide. Online Resource 2 contains the numerical data, reconstruction script and existing case notes supporting Tables 1–6 and Figures 3–5. The case notes also document the examples in Figure 1. These resources add no new experiments or inferential tests.

References

- [1] Blondel M, Fujino A, Ueda N, et al (2016) Higher-order factorization machines. In: Advances in Neural Information Processing Systems, URL <https://proceedings.neurips.cc/paper/2016/hash/158fc2ddd52ec2cf54d3c161f2dd6517-Abstract.html>
- [2] Carbonell J, Goldstein J (1998) The use of MMR, diversity-based reranking for reordering documents and producing summaries. In: Proceedings of the 21st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval, pp 335–336, <https://doi.org/10.1145/290941.291025>, URL <https://dl.acm.org/doi/10.1145/290941.291025>
- [3] Gerber J, Saxer J, Kreiner B, et al (2026) Benchmarking state of the art website embedding methods for effective processing and analysis in the public sector. *Journal of Intelligent Information Systems* 64:97–120. <https://doi.org/10.1007/s10844-025-00954-4>, first published online 13 June 2025
- [4] Guo T, Wang C, Liu Y, et al (2025) Leveraging inter-chunk interactions for enhanced retrieval in large language model-based question answering. *Neurocomputing* 650:130931. <https://doi.org/10.1016/j.neucom.2025.130931>

- [5] Karpukhin V, Oguz B, Min S, et al (2020) Dense passage retrieval for open-domain question answering. In: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), pp 6769–6781, <https://doi.org/10.18653/v1/2020.emnlp-main.550>, URL <https://aclanthology.org/2020.emnlp-main.550/>
- [6] Lee D, Jo Y, Park H, et al (2025) Shifting from ranking to set selection for retrieval augmented generation. In: Proceedings of ACL, pp 17606–17619, <https://doi.org/10.18653/v1/2025.acl-long.861>
- [7] Lin H, Bilmes J (2011) A class of submodular functions for document summarization. In: Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies, pp 510–520, URL <https://aclanthology.org/P11-1052/>
- [8] Qwen Team (2024) Qwen2.5: A party of foundation models. URL <https://qwenlm.github.io/blog/qwen2.5/>, official model release
- [9] Ram P, Gray AG (2012) Maximum inner-product search using cone trees. In: Proceedings of the 18th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp 931–939, <https://doi.org/10.1145/2339530.2339677>
- [10] Tian F, Ganguly D, Macdonald C (2025) Is relevance propagated from retriever to generator in RAG? In: Advances in Information Retrieval, pp 32–48, https://doi.org/10.1007/978-3-031-88708-6_3
- [11] Trabucco B, Kumar A, Geng X, et al (2021) Conservative objective models for effective offline model-based optimization. In: Proceedings of ICML, pp 10358–10368, URL <https://proceedings.mlr.press/v139/trabucco21a.html>
- [12] Trivedi H, Balasubramanian N, Khot T, et al (2022) MuSiQue: Multihop questions via single-hop question composition. Transactions of the Association for Computational Linguistics 10:539–554. https://doi.org/10.1162/tacl_a_00475, URL <https://aclanthology.org/2022.tacl-1.31/>
- [13] Wiseman S, Rush AM (2016) Sequence-to-sequence learning as beam-search optimization. In: Proceedings of EMNLP, pp 1296–1306, <https://doi.org/10.18653/v1/D16-1137>
- [14] Xiao S, Liu Z, Zhang P, et al (2024) C-Pack: Packed resources for general chinese embeddings. In: Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval, pp 641–649, <https://doi.org/10.1145/3626772.3657878>
- [15] Xiong W, Li XL, Iyer S, et al (2021) Answering complex open-domain questions with multi-hop dense retrieval. In: International Conference on Learning Representations, URL <https://openreview.net/forum?id=EMHoBG0avc1>

- [16] Xu F, Shi W, Choi E (2024) RECOMP: Improving retrieval-augmented LMs with compression and selective augmentation. In: International Conference on Learning Representations, URL <https://openreview.net/forum?id=mlJLVigNHp>
- [17] Yang Z, Qi P, Zhang S, et al (2018) HotpotQA: A dataset for diverse, explainable multi-hop question answering. In: Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing, pp 2369–2380, <https://doi.org/10.18653/v1/D18-1259>, URL <https://aclanthology.org/D18-1259/>
- [18] Zaheer M, Kottur S, Ravanbakhsh S, et al (2017) Deep sets. In: Advances in Neural Information Processing Systems, URL <https://proceedings.neurips.cc/paper/2017/hash/f22e4747da1aa27e363d86d40ff442fe-Abstract.html>
- [19] Zhai F, Tang W, Jin J (2025) RAG-LER: Ranking adapted generation with language-model enabled regulation. Neurocomputing 656:131514. <https://doi.org/10.1016/j.neucom.2025.131514>
- [20] Zhang J, Zhang H, Zhang D, et al (2024) End-to-end beam retrieval for multi-hop question answering. In: Proceedings of NAACL-HLT, pp 1718–1731, <https://doi.org/10.18653/v1/2024.naacl-long.96>
- [21] Zoraliloglu Y, Yalcin E (2026) Dynamic feedback loops in recommender systems: Analyzing fairness, popularity bias, and user group disparities. Journal of Intelligent Information Systems 64:2091–2119. <https://doi.org/10.1007/s10844-026-01025-y>